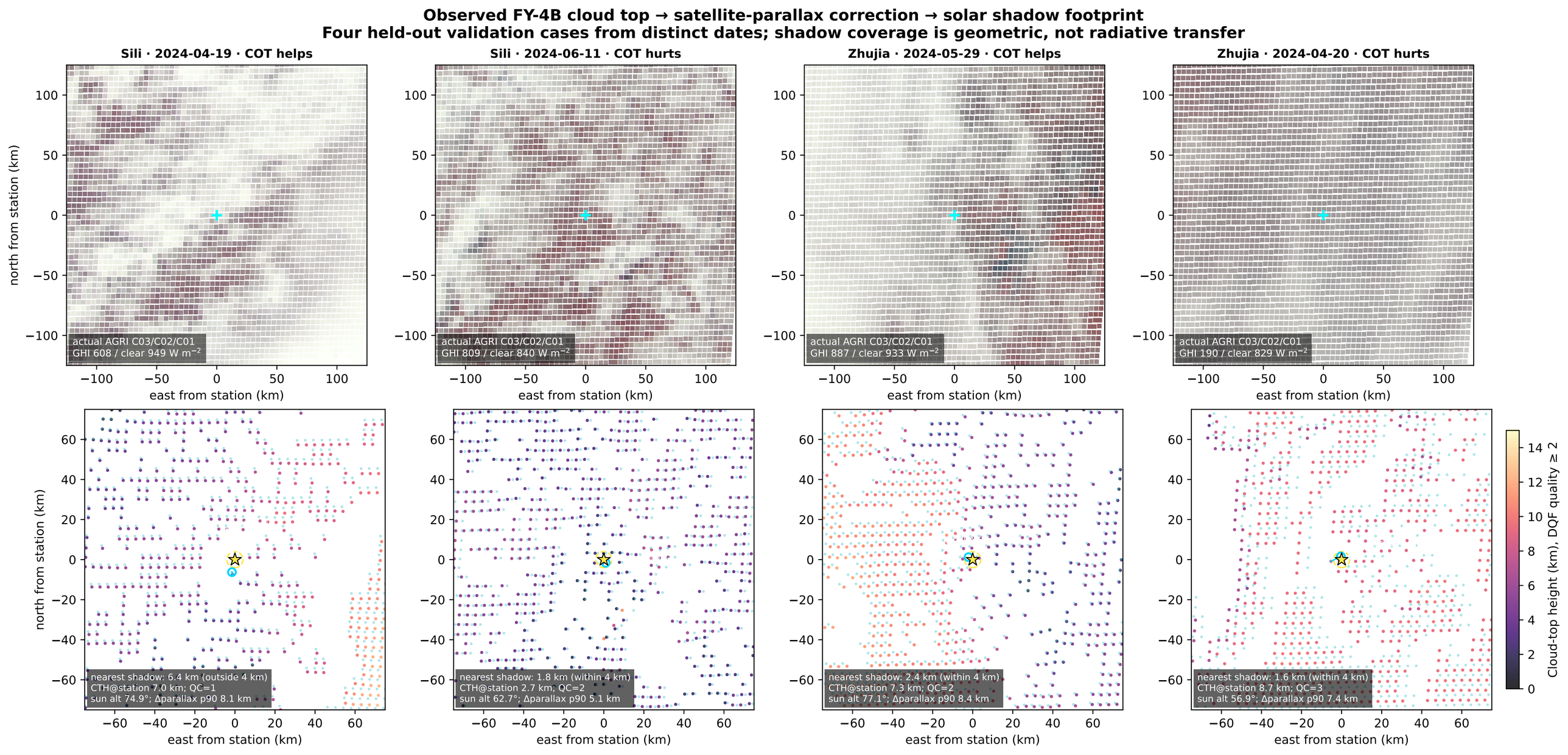

Observed FY-4B cloud top → satellite-parallax correction → solar shadow footprint
Four held-out validation cases from distinct dates; shadow coverage is geometric, not radiative transfer
Sili · 2024-04-19 · COT helps
Sili · 2024-06-11 · COT hurts
Zhujia · 2024-05-29 · COT helps
Zhujia · 2024-04-20 · COT hurts
north from station (km)
east from station (km)
actual AGRI C03/C02/C01
GHI 608 / clear 949 W m⁻²
GHI 809 / clear 840 W m⁻²
GHI 887 / clear 933 W m⁻²
GHI 190 / clear 829 W m⁻²
nearest shadow: 6.4 km (outside 4 km)
CTH@station 7.0 km; QC=1
sun alt 74.9°; Δparallax p90 8.1 km
nearest shadow: 1.8 km (within 4 km)
CTH@station 2.7 km; QC=2
sun alt 62.7°; Δparallax p90 5.1 km
nearest shadow: 2.4 km (within 4 km)
CTH@station 7.3 km; QC=2
sun alt 77.1°; Δparallax p90 8.4 km
nearest shadow: 1.6 km (within 4 km)
CTH@station 8.7 km; QC=3
sun alt 56.9°; Δparallax p90 7.4 km
Cloud-top height (km), DQF quality ≥ 2